

General Instructions:

- This question paper consists of 6 sections with 19 questions (as this is the 19th edition of Technothon).
 - The marking scheme is given below, one of the unique schemes you might ever experience.
 - There is one section (Section 5) where you have to select the marking scheme according to your choice in the first question (Q.0) of the section.
 - Failing to opt for the marking scheme of your choice will result in selecting the marking scheme in the first option by default.
 - No ambiguity related to the marking scheme will be entertained later.
 - If any discrepancy in the paper is found, the final decision regarding the same will be taken by Team Technothon.
 - Integer type answers can have any integer value (NOT restricted to 0 to 9).
 - The Marking scheme given for the particular question shall not be considered for evaluation and the marking scheme given in the section shall be considered for evaluation.
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- Please ignore the auto-generated leader board. You will get the prelims results on the Technothon home page, where you registered for the test.

Selection Criterion and Result:

- The ranking will be based on the total marks obtained in all the sections.
- The result will be declared on or before August 8th, 2022 on our website technothon.techniche.org.in.
- To check your result, log in with the Roll number and password provided in your admit card.
- The top 50 teams will be invited to IIT Guwahati for the Mains and will be awarded Gold certificates.
- The next 200 will be awarded Silver certificates.

Section 1

1. Which kid does not dream of a magic gimmick that makes him a superhero? It is what happened to Ben Tennyson, a 10-year-old boy - when he came into possession of a mysterious watch, the 'Omnitrix', that could transform him into ten superhero aliens.

Ben is a generous, curious, and bold boy who can not compete against the oppression of school bullies because of his slim physique. It was the last day of school, and Ben was eagerly waiting for the vacation to begin as he was going to go summer camping with his grandfather Max, a man who is enterprising, energetic, and loving. After finishing the exam, Ben comes running out of the school gate and hugs his Grandpa, Max Tennyson.

To his surprise, Max has also brought Ben's cousin, Gwen, a clever, smart, and funny girl, and said she too would come with them. Ben and Gwen can never stop fighting, and they both never miss a chance to tease and make fun of

each other. The moment Ben sees her, he makes faces at her; Gwen too teases him. Max scolds them both, and all the three Tennysons get into Max's Recreational Vehicle (RV), which he calls the 'Rust Bucket.' It is outfitted with very advanced technology, including jet propulsion and laser weaponry with private satellite connectivity and begin their journey not knowing what would come on their way.

Main Characters

Ben

Ben was just an average boy until he found the Omnitrix, an alien gadget that allowed him to harness the genetic codes contained within to morph into several various extraterrestrial species. Although highly self-centered and arrogant at times, with a tendency to throw himself into situations without properly understanding the consequences.

Gwen

Gwen uses magic by casting spells and reciting spoken incantations thus enabling her to use various Her quick thinking and good instincts give her an edge in dealing with dangerous situations. magical powers and abilities, However, she's sometimes too trusting with people who want to exploit her goodwill, leading her (or others) into harm's way.

Max

Ben and Gwen's grandfather take the two on a cross-country road trip for summer vacation. Later, Max reveals he was a part of the Plumber's intergalactic police force and handled many extraterrestrial encounters in the past. His history as a plumber has given him knowledge, training, and equipment for fighting aliens as well as an unconventional taste in food.

Vera

Vera Tennyson is the older sister of Max and the grandaunt of Ben and Gwen Tennyson. Vera is extremely overbearing with her family and has a habit of unintentionally causing undue stress on others with her good intentions.

Eddie Grandsmith

Eddie tends to be arrogant. When he first met [Ben](#) and [Gwen](#), he was rude and off-putting. After his [grandfather](#) convinced him to give a tour for them, he mentioned how good a swimmer he is, causing Ben to mock him. He strongly dislikes how his family solves every problem with money.

Donovan Grandsmith

Donovan was one of [Max Tennyson](#)'s friends. When they were younger, Donovan gained a large amount of money. As a result, he decided to travel the world. Many years later, Donovan became a billionaire and started a chain of hotels.

Azmuth

Azmuth is super-intelligent, despite his small size. Along with his intelligence, Azmuth is gifted with years of experience and knowledge gained thanks to his age. Azmuth is a skilled inventor having created numerous inventions his greatest being the Omnitrix.

Aliens

CannonBolt

When rolled up, Cannonbolt becomes like a motorized wheel that can move at great speed and can increase his rotational momentum to slam into and/or run over enemies. Between that and his excellent hearing, Cannonbolt receives something akin to echolocation - giving him "eyeless" vision.

Diamondhead

Diamondhead is capable of encasing enemies in crystals and can create crystals inside of objects and cause them to spike out. He can manipulate the atomic structure of his physiology at will, allowing him to form his limbs into diamond weapons such as blades and bludgeons.

Ripjaws

Ripjaws was [an aquatic alien](#), with a design resembling a humanoid angler fish. When in water, his legs would shift into a mermaid-like tail. Ripjaws had one of the most fearsome alien designs in the series. However, he can't breathe outside the water with gills which is a major drawback

Ghostfreak

Ghostfreak's powers are composed of traditional ghosts flight, levitation, phasing through solid matter, intangibility, invisibility, and body possession. Ghostfreak possesses enhanced strength, as shown when he used his tentacles to grab a bus full of people.

Wildmutt

As he has no eyes, his "vision" is made up of a dull 3D mapping of where his senses originate, similar to a thermograph. Combined with his enhanced hearing, his senses take the form of a radar or sonar. Wildmutt can sense invisible entities, therefore making invisibility useless against him.

Grey Matter

Grey Matter's small size belies his great calculative and technical intellect. He can create complicated machines from spare parts, understand the function of any device, and can sometimes manipulate living things by stimulating certain parts of their nervous system if such beings can be controlled by external stimuli.

FourArms

Four Arms' most prominent ability is his enhanced strength, being able to lift several times his weight and punch through various resilient materials with ease. Four Arms can create strong shockwaves by clapping all four of his hands together. Alternatively, he can create shockwaves by slamming his fists on the ground.

Stinkfly

Stinkfly possesses two insectoid wings that allow quick and agile flying and can change directions at exceptional paces. Stinkfly can excrete high pressured streams of liquids, they can be both a toxin or an adhesive, instinctively choosing what type of slime it can be. Besides liquids, Stinkfly can release poisonous and methane gas.

HeatBlast

Heatblast is pyrokinetic, meaning, he can generate and manipulate intense heat and fire from his body. He can form fire into any shape that he chooses, commonly fireballs. To top it all off, he can even absorb heat and fire from external sources, allowing him to put out flames in short amounts of time.

Upgrade

The nanites that make up his body allow Upgrade to reshape and move like a liquid even though he's a solid being. His physiology also makes him extremely flexible. Upgrade's DNA is "programmed" with blueprints of every alien technology, allowing him to merge with and possess any technology by encasing himself over it.

XLR8

Using the orbs on his feet, XLR8 can reach running speeds of 500 miles per hour within two seconds. He can accelerate so quickly that time appears completely stop to him, allowing him to change the positions of things around him in mid-motion, Even without his orbs, he is naturally capable of thinking, acting, and reacting at superhuman speed.

Villains

Vilgax

A warlord regarded as one of the most feared aliens in the galaxy, who stops at nothing to take the Omnitrix, which he wishes to use to create a transforming super army. In the past, Vilgax was an enemy of Max. he was cybernetically enhanced, making him a match for every alien Ben could access at that time, therefore becoming the boy's archenemy.

On a scale of 1 to 10, how excited are you for Technothon? **(Integer Question)**

SECTION-2: Finding the Omnitrix

Marking Scheme - Power sequence

Power Scheme: With negative marking

For consecutive correct questions, you will be awarded 21, 22, 23 and so on.

However, if you leave a question or answer it incorrectly, the sequence is broken, and you will start

again from 2.

If you answer the question incorrectly your marks will be deducted in the same pattern but different sequence

E.g., Solving 3 questions,

For all correct you get $2+4+8 = 14$

For all incorrect you get $-1-2-4 = -7$

For RRW you get $2+4-1 = 5$

For RWR you get $2-1+2 = 3$

For WRW you get $-1+2-1 = 0$

For RRU you get $2+4+0=6$

And so on.

(R -> right answer, W-> wrong answer, U-> Unattempted)

Maximum marks : 14

Minimum marks : -7

2. Max stops his RV near a forest for taking a small break. As he was driving the whole day, he asked the kids to watch out for the van while he took a small nap. Ben and Gwen nod obediently in front of him; then get down from the van and start arguing about who should be on the lookout. While fighting, Gwen notices a tall tree which fell nearby, with a creeper surrounding it. She knows Ben is too bad at math and too lazy to do any sort of calculations. So, she cleverly devises a plan. She immediately gets a measuring tape from the van and measures the dimensions of the tree. "The tree has a height of 24 feet and a circumference of 3 feet. It is surrounded by a spiral creeper uniformly encircling it exactly six times from start to end of the tree. What is the length of the Creeper? If you can find the length faster than me, then I'll be on the watch out; if not, then you should", Gwen says. Ben agrees.



What is the length of the creeper?

- a. Greater than and equal to 25 and less than 30
- b. Greater than and equal to 30 and less than 35

- c. Greater than and equal to 35 and less than 40
- d. Greater than and equal to 40

3. Gwen's plan worked as expected; She found the answer faster than Ben, leaving him to guard the van while she went to sleep. Ben sits outside the van and plans to take revenge on Gwen. He suddenly sees a meteor and mistakes it for a shooting star, but it changes its course and crashes nearby. Inside the crater, he finds a shell with fumes coming from it. Ben goes near it to check it out. He finds some letters inscribed on the metal surface of the shell and presumes it to be a puzzle.

He also finds a display glass with "Enter Password" on it, and a keyboard with numbers below it. There was also a hint given below: Maximum count of 'ITS HERO TIME'. Ben figures out that he has to find the maximum number of ways to get ITS HERO TIME from the puzzle inscribed on the shell, starting from the letter I at the top and to the letter E at the bottom. The maximum number would be the password to open the shell.



Find the password and help Ben unleash the secrets of the fuming shell.

- a. 248
- b. 250
- c. 252
- d. 254

4. Ben correctly enters the password, and the shell opens. Inside the shell appears a strange machine similar to a wristwatch - **The Omnitrix**, the item sought by Vilgax, an intergalactic alien warlord and conqueror, but little did Ben know about it. Driven by curiosity, Ben goes closer to it and tries to touch it; the Omnitrix jumps onto his wrist, unwillingly merging with his DNA.

Ben tries in every way to take it off, but fails, then presses a key on that strange device, which displays some figures. Selecting one of them, he transformed into an alien covered with lava. Ben, despite the initial shock, realizes that it has superpowers - it can throw balls of fire. But because of his clumsy experiments, he sets the whole forest on fire. Seeing the fire, Grandpa Max and Gwen head into the forest, worried about Ben. But they find out that he is the reason for the fire. Fortunately, Max had some fire extinguishers in his RV to save the day, but they were mixed with similar-looking oxygen cylinders.

There are 5 containers of 17 cylinders each. The Oxygen cylinders weigh 1 kg each, and fire extinguishers weigh 2 kg each. Each container has either oxygen cylinders (has only oxygen cylinders) or fire extinguishers (has only fire extinguishers). As the fire was spreading fast, they had little time to segregate the containers. Given a weighing machine, devise a method to differentiate and determine all the fire extinguishers with just one weighing. Assume that any number of cylinders from any number of containers can be placed on the weighing machine.

Find the minimum value of the total number of cylinders that can be placed on the weighing machine.



INTEGER QUESTION

SECTION - 3: Dr. Animo's Pursuit

Marking Scheme - Affinity in a row

Q1, Q2, and Q3 have marks 2, 4, and 6, respectively **BUT**

There are two conditions:

- a. There would be no reduction in the marks of this section if you solve 2 consecutive questions correctly, also

b. If the above criteria are not met alongside you solving a question incorrectly, the total marks you have scored in this section are halved.

For example :

For a 3 question section-

If you solve Q2 and Q3 correctly but Q1 incorrectly or leave it unattempted then you will get $0 + 4 + 6 = 10$ marks.

If you solve Q1 and Q3 correctly but Q2 incorrectly or you have not attempted it, then you will get $(2 + 0 + 6)/2 = 4$ marks.

Similar rules are applied for a section containing 4 questions with the last question having 8 marks.

Maximum marks : 12

Minimum marks : 0

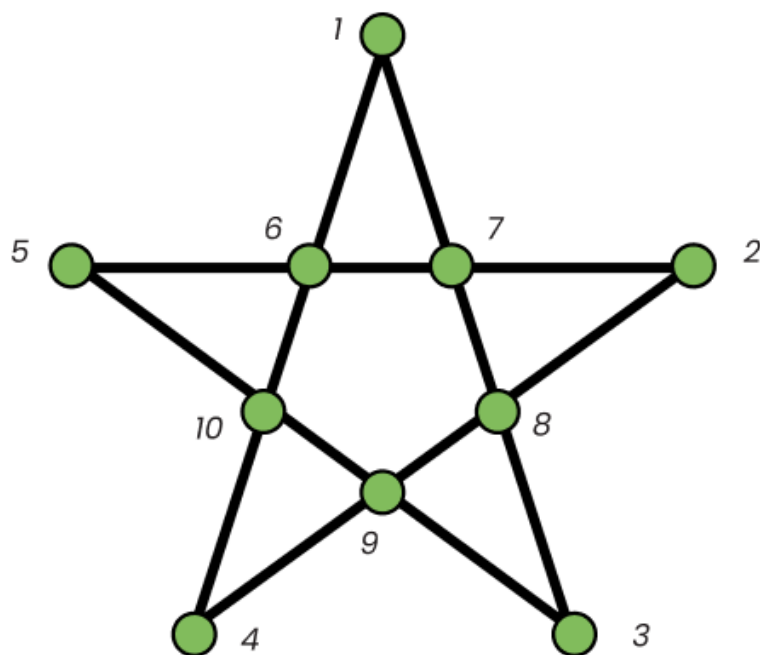
5. Dr. Animo was once a respected and promising researcher in veterinary science. Still, his career was cut short when it was discovered that he was performing twisted genetic experiments on animals. He had hoped that his research would win him the Verities Award, which is an award in veterinary science. Because of the nature of his research, he lost the award to another doctor named [Kelly](#). So, he decides to break into the venue and steal the award before the award-ceremony day.

The Award was kept under protection inside a giant star with a high voltage current passing through it. Ten switches are placed on the intersecting points of the stars, one at every point, as shown in the image below.

He can start from any point (including the points of intersection of any two lines) and then make one long jump over to another point, directly jumping to the third one in a row. (For example, in the fig given, if he starts from Point 1, then he can go to vertices 8 and 10 but not to 6 and 7.) After reaching a vertex, he should turn off the switch. Similarly, in the next turn, he can again start from any point, apart from those already switched-off, and continue jumping to the third point. Note that he can jump over a switched-off point but cannot start from it. And he can only traverse through straight lines. He can only take the award safely if the maximum number of switches are turned off.

Animo uses his mutated giant frog to take the big leaps/jumps. If he disobeys any of the rules mentioned above, then the alarms would ring and alert the security, making it impossible for him to steal the award.

Find the maximum number of switches Dr Animo was able to turn off.



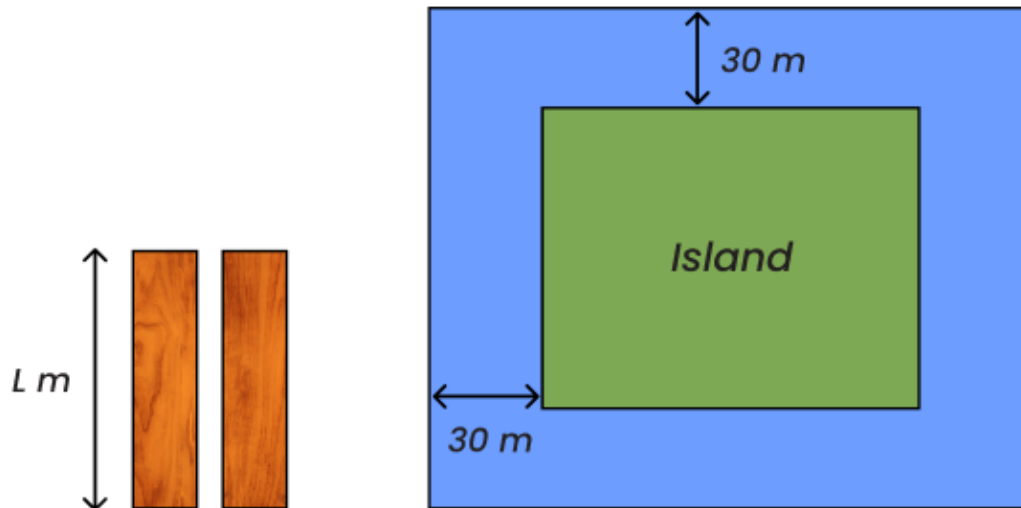
INTEGER QUESTION

6. Animo successfully took the award and left to his laboratory. The Tennysons were passing through the same road when they saw the security hustling and red alarms beeping. When enquired, they came to know Dr. Animo had stolen the award and had taken it to his lab on the city outskirts. They immediately head toward Dr. Animo's lab.

After reaching the lab, they all start attacking Animo. But feeling scared that Max and Gwen would get hurt, Ben tells them to find and release the trapped animals while he fights Animo with his new superpowers.

Max and Gwen come to know that Animo has an island where he trapped and caged many innocent animals to do restricted and classified experiments on them. So, they head towards Animo's artificial island, built at the back of his laboratory to save the animals. It is surrounded by a rectangular moat that measures 30m in width, as shown in the image below.

They find two planks of 'L' m each, but there's no way they can be nailed together. So, Gwen needs to comprehend a way to cross the moat from outside and reach Animo's island using the two planks. What should be the minimum value of L that can be used to cross the moat?



- A. Less than 15
- B. Greater than and equal to 15 but less than 20
- C. Greater than and equal to 20 but less than 25
- D. Greater than and equal to 25 but less than 30

7. Max and Gwen go to the island and successfully save the animals. Meanwhile, Ben changes into Cannonbolt and starts destroying Animo's lab. Cannonbolt is an alien with yellow limbs, which can become like a motorized wheel that can move at great speed and can increase his rotational momentum to slam into and run over enemies. Animo tries all possible ways to stop him, but he won't stop and starts hitting Animo. Dr. Animo, being hurt already, runs to save himself.

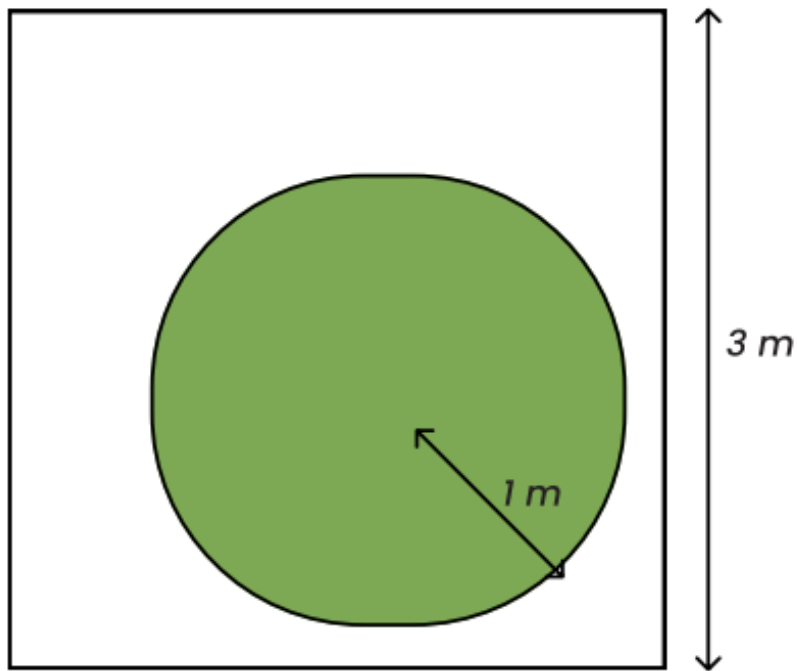
He sits on his giant frog and takes a big leap into his secret tunnel that would lead him to his underground hideout. Ben reaches the site of the tunnel to find a big square with a circular hole in it. He figures out that each side of the square is 3 m and the diameter of the circular pit is 2 m. The sides of the square had current passing through; so if

Ben touched any of the sides, he will get badly hurt. So, Cannonbolt should jump and only pass through the circular pit without touching the sides of the square. What is the chance that Cannonbolt would safely roll into the tunnel?

Type the answer in the following format- (num+den)*(num-den)

For example, if the chance is m/n , then the answer should be - $(m+n)*(m-n)$

INTEGER QUESTION



Ben goes into the tunnel, fights with Animo and retrieves the award that Animo has stolen. They return the award to the council of veterinary science and hand the rescued animals over to them.

SECTION - 4 : At the Bermuda triangle

Middle benefits

The typical marking of the section will be +3 +2 +3 for the consecutive questions in the section. If you attempt the middle question correctly, then you will have some benefit and marks of the other 2 will increase as +5 +2 +5. The negative marks will be -1 in each case.

Maximum marks : 12

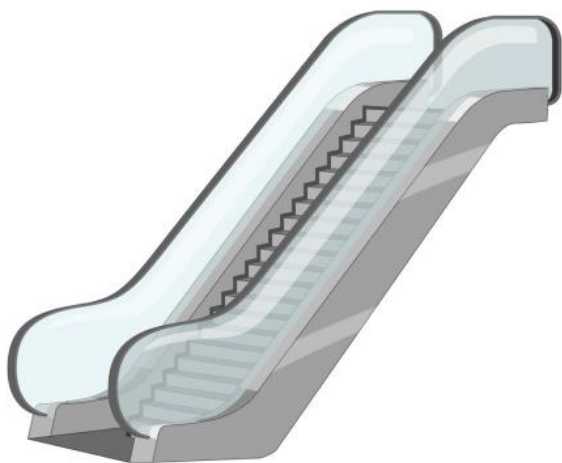
Minimum marks : -3

8. Ben and Gwen insist Max take them to someplace new and exciting. Max remembers that there is an inauguration ceremony of an underwater resort in the Bermuda Triangle, owned by his old friend, Donovan GrandSmith, So he decides to drop by there. At the entrance, they receive a warm welcome from Donovan himself but also meet his grandson, Eddie, who doesn't seem to be excited about their guests. Donovan gives them a personnel tour of the resort and guides them to his giant submarine, 'The Undersea Manta-Ray.'

Max and Donovan recollect their past fun memories. Donovan reminds Max about the "Round robin tournament" they used to play. The kids became curious when they heard about it and asked if they could play the game too. Max and Donovan laugh and agree to play a game. Donovan says he would be the judge leaving Max, Ben, Gwen, and Eddie to play. All four of them play separate games with the other three. The winner of each game gets 4 points, and if the game ends up with a draw, each player gets 2 points. Eddie, being a pro in the game as his grandfather, expectedly wins the league with 10 points. Gwen scored 6 points and 5 goals, while Ben has 2 points. Total number of goals was 11. In a match between Ben and Gwen, Gwen defeats Ben with a score of 2:1. Find the total number of goals scored by Eddie in the game against Ben.

INTEGER QUESTION

9. They had to take an escalator after entering the submarine. Donovan said, his face beaming with pride, that it is the largest escalator built so far. "Whoa, impressive, Mr. GrandSmith!," Gwen says, "But how many steps does it have?". Donovan replies, "I'll leave it for you to find out." He continues, "Suppose that Ben and Eddie are descending a similar escalator. Eddie walks down, taking 250 steps to reach the bottom; Ben ran down, taking 375. If I tell you that Ben ran three steps in the time it took Eddie to walk one, find out how many steps will be visible when the escalator is stopped?"



Gwen, being a math nerd that she is, opens her pocketbook and scribbles something on the paper and tells the answer before getting down the escalator. What is the answer that Gwen told?

- A. 400
- B. 500
- C. 700
- D. 800

10. Everyone, except Ben (of course!), gets impressed by her math skills and speed. Gwen smiles with triumphant makes a face at Ben. Ben ignores her.

As soon as they all enter the Manta Ray, Donovan starts the engine. As Eddie was standing closer, the sudden and heavy thrust of the engine made him jerk away, and he slipped down into a cylindrical tank but managed to cling to a rod. Ben goes behind the door and transforms into Stinkfly.

The cylindrical tank is 200 m high and has a circumference of 300 m. Eddie is hanging on a rod which is 50 m from the top. Stinkfly is outside the tank, 50m up from the bottom, but diametrically opposite the rod. But due to the high Figure out the shortest path which Stinkfly can use to reach the rod to save Eddie, and find out exactly how far does he fly ? (Ignore the thickness of the glass.)

After saving Eddie, Ben changes back from Stinkfly to his normal self and joins the others. Max says, “Well, I guess someone deserves to be given credit for saving the day. Right, Ben?” and winks at him. Ben smiles and makes a face at Gwen. Gwen frowns.

INTEGER QUESTION

SEC - 5 : A visit to Aunt Vera’s

Marking Scheme - Bitonic Sequence

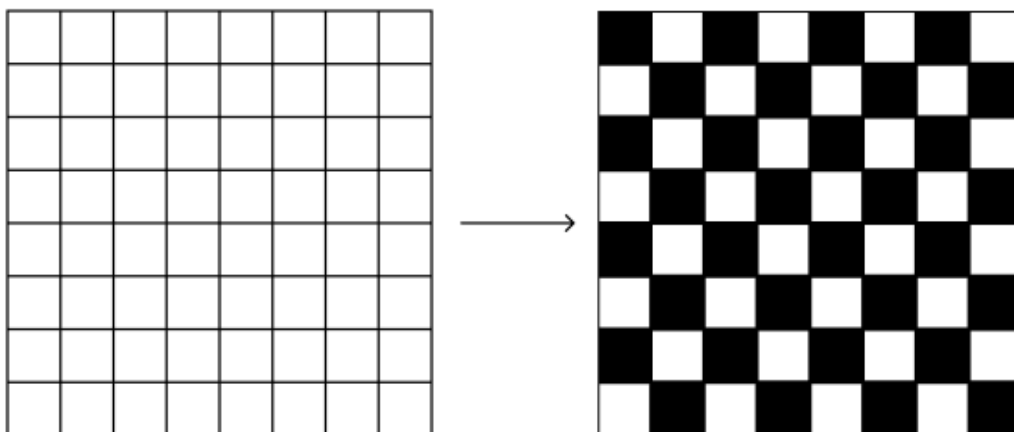
Bitonic sequence are those which are strictly increasing/decreasing in the start and then strictly decreasing/increasing. The marking scheme will be +4,+2,+4 for question 1,2,3 respectively in the section with a negative of -2,-1,-2.

Maximum marks : 10

Minimum marks : -5

11. Max takes Ben and Gwen all the way to a retirement centre to visit their Aunt Vera, but when they reach her house, they find many containers with paint on the lawn. After exchanging pleasantries, Aunt Vera tells them that the Mayor of the town has announced a huge cash prize for those who would paint the floors of his newly built office in the shortest time. She tells them she wanted to win the prize, so she kept the required paint at ready and was about to leave. But Ben and Gwen, being the good kids they are, offer her help and tell them they would paint the floor instead of her.

The office had a floor plan which mimicked an 8*8 chessboard, with each of the 64 rooms having a door in each of its four walls. Initially, all the floors in all the rooms were painted white. Then the king ordered the floors to be repainted so that they alternated like the squares of a chessboard. To do this, the painter has to walk through the palace, repainting floors in all the rooms he visited from white to black and black to white. The painter was allowed to exit the palace and re-enter it at another door. Gwen quickly devises a plan to finish the task faster, and Ben changes to XLR8 to paint the floor. What is the minimum number of times he has to paint the floors of the rooms to make them look like a chess board?

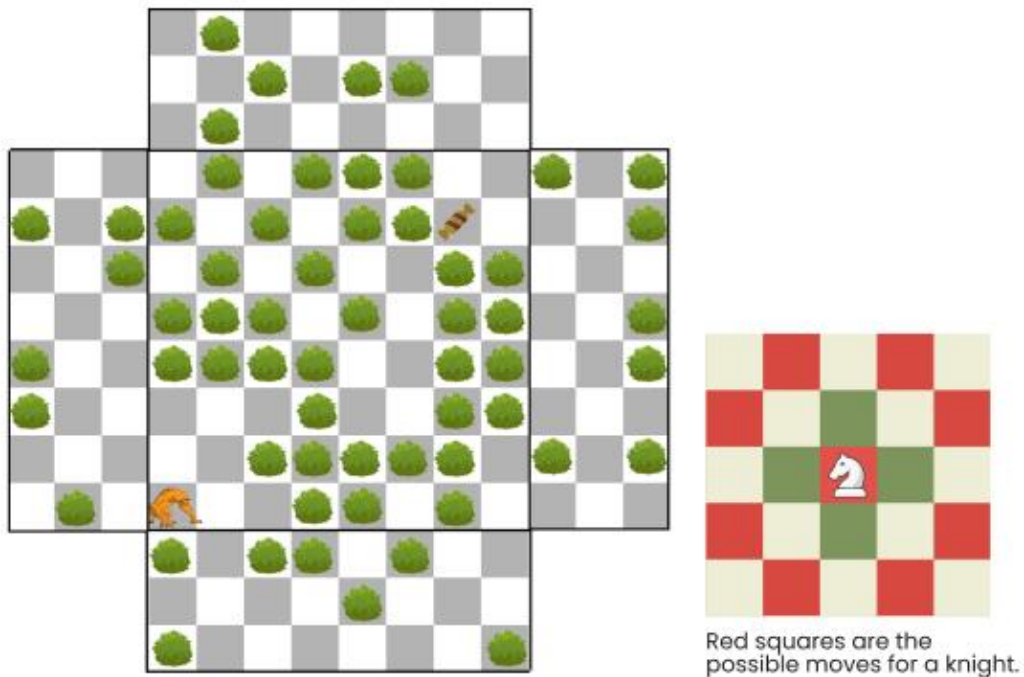


- A. 40
- B. 60

- C. 80
- D. NONE OF THE ABOVE

12) Ben and Gwen together not only finished the work but also won the prize and gave it to Aunt Vera, leaving her so happy and impressed. Aunt Vera makes the best chocolate. As a token of affection and triumph of the kids, after dinner, Aunt vera brings a big jar of chocolates and gives them to everyone. Ben still craves for the chocolate, even after eating a jar of it. Ben pleads with her to give more, but she strictly denies saying he might get sick, and goes away with the jar. Ben knows that Aunt Vera hides her chocolate in a hidden storeroom in her garden, and one has to pass through a secret maze to reach it. But he never dared to go there before. Now that he has the Omnitrix, Ben grins slyly and waits until night falls.

Once everyone's asleep, Ben tiptoes to the garden, and BOOM! he turns into Wildmutt and smells the chocolate. The maze to the store room is as shown below.

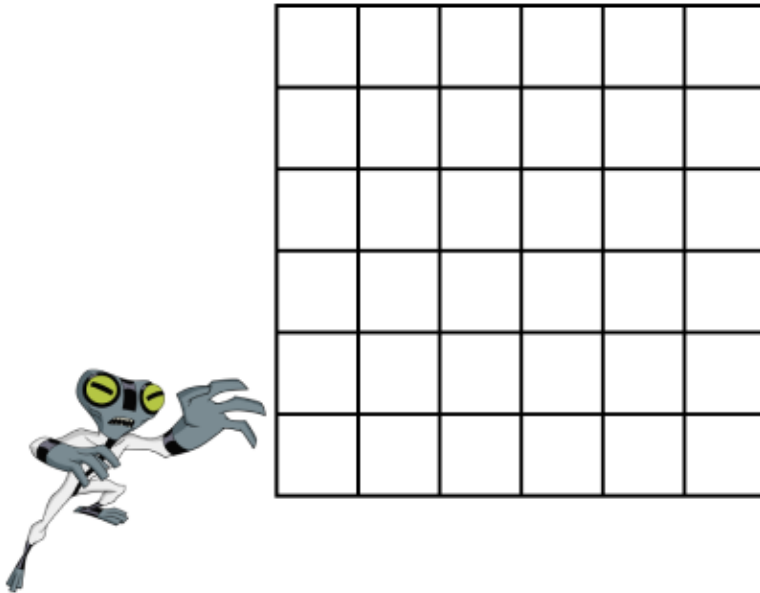


As it was dark, ben mistakenly unlocks the 'Knight mode' of the Omnitrix. When this mode is on, the transformed aliens can only move as a knight(horse) on the chessboard. The red dots indicate the locations of Wildmutt and the store room. The black blocks have thorny bushes, and Ben cannot step on it. Now, figure out the minimum number of moves Wildmutt has to take to reach the chocolate.

- A. 10
- B. 12
- C. 14

D. None of the above

13) Ben reaches the storeroom, finishes off all the chocolate and returns to his room and sleeps like nothing happened. The next day morning, his stomach hurts because of the over-eating of the chocolate, and Ben decides to take a walk in the nearby park. He encounters an odd-looking old man and senses that something is fishy about him. He secretly follows him inside a burrow leading to an underground compartment. He overhears the conversation between the old man and his pals and figures out that they are the local gangsters who kidnap kids and make money out of them. Ben peeped inside and saw some kids locked inside cells. Ben knew that it was 'Hero Time'; he turns the Omnitrix and transforms into Greymatter.



The kids were locked in 36 cells, and Ben wants to rescue them all. But the 'Knight mode' is still ON, so he can only move as a knight(horse) on the chessboard. Ben can start moving from any cell, but he must ensure that he goes to all the cells and unlocks all the doors. Find the minimum number of moves in which ben can rescue all the kids.

INTEGER QUESTION

SEC - 6 : Preparations for war

Marking Scheme - Choose your ally

The most interesting marking scheme you might have seen. Out of the given two options; choose the marking scheme that you think is best for you. (You can check the questions before marking your choice)

Maximum marks : 20

Minimum marks : -10

Affinity in a row

Arithmetic game

You will be awarded 2,4,6,8 for consecutive correct answers and a negative of -1,-2,-3,-4 for incorrect so

For example:

If Your sequence is RRRW, then your score is $2+4+6-4 = 8$

But if your sequence is RWRR, then your score would be $2-2+6+8 = 14$

And if your sequence is WRUR, then your score is $-1+4+0+8 = 11$

Where (R -> right answer, W-> wrong answer, U-> Unattempted)

Question 0) Choose between the two marking scheme.

- A. **Arithmetic Game**
- B. **Affinity in the Row**

14) After a day well spent at Aunt Vera's, the Tennysons bid her goodbye and get back on their adventurous road trip. Ben sleeps sound fully, as he had a long day. But he gets a vision in which Vilgax, the evil alien warlord, threatens him that he will soon come to earth to take the Omnitrix back. Ben tells Max about his vision which worries him. When Ben and Gwen ask the reason, Max tells them about his 'secret past'; that he was a deputised agent of "The Plumbers," a secret government organisation that dealt with extraterrestrial phenomena, which was disbanded after Max defeated Vilgax himself for the first time.

Ben and Gwen keep staring at him, their mouths wide open. Max tells that Ben's life might be in danger and that he should be trained appropriately on how to use the Omnitrix. Max knows Azmuth is the correct person to do it. Azmuth is a Galvan from the planet Galvan Prime. He is the creator of the Omnitrix and also a trusted advisor to the Plumbers. Max sends a message to Azmuth to save his grandson and also Earth from Vilgax and his army. Azmuth had huge respect for Max and trusted him a lot. So he agrees to come to Earth and hence, starts making preparations for his journey.

It takes ten days to reach Earth from the Galvan Prime through his spacecraft. But Azmuth's space crafts can only carry fuel that can suffice for a journey of six days. So, Azmuth decides to take some of his assistants to accommodate fuel for his journey. Everyone pilot their own spacecraft and the fuel can be transferred anytime between any of the spacecrafts. With how few assistants can Azmuth successfully reach Earth?



- A. less than 3
- B. 3
- C. greater than 3
- D. not possible with any number of assistants

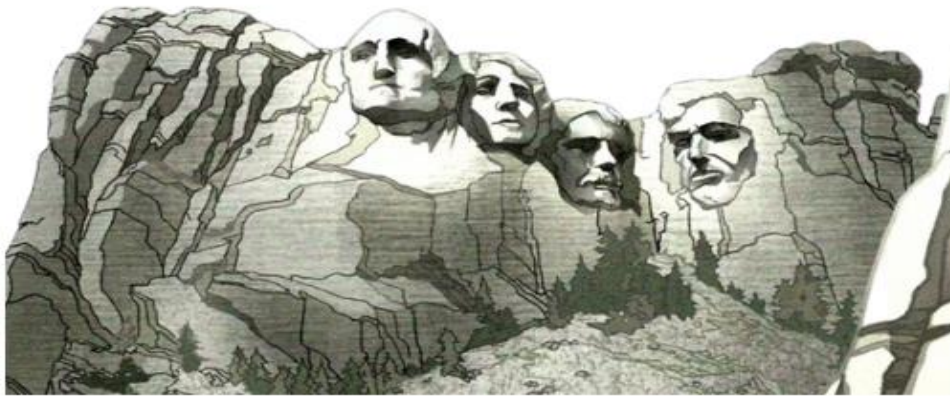
15) Azmuth comes to Earth, meets Max and knows the complete scenario. He turns off the 'Knight Mode' and unlocks the Master Control of the Omnitrix. He trains Ben on how to use it and gives some tactics on how to win over Vilgax. Azmuth urges Max that they have to retrieve the weapons needed from the secret base of "The Plumbers" in Mount Rushmore before Vilgax comes looking for him.

The three Tennysons reach Mt. Rushmore and enter a tunnel behind the rocks, leading them to a tall, old building with 37 floors. Max realises that he has forgotten the password needed to get a direct entry to the armoury unit. The only other way possible was to go in a specially designed elevator.

If someone wants to go up, the elevator moves up by five floors if it can. If the elevator cannot move up by five floors, it stays in the same spot (if you are on floor 33 and press up, the elevator stays on floor 33). And if someone wants to go down, the elevator moves down by eight floors if it can. If it cannot, then the elevator stays in the same spot. (if you press down from floor 8, the elevator stays on floor 8). The elevator starts on floor 1. Max doesn't remember the floor on which the armoury unit was present; he only remembers that it is on the floor, which takes the maximum number of stops when starting from floor 1.

Gwen does the math in a snap and finds the required floor number.

Considering they try to go to every floor optimally, which floor will take the maximum number of stops?



- a. 33
- b. 34
- c. 35
- d. 37

16) Vilgax decides to come to Earth to retrieve the Omnitrix; so he wants to keep his army ready. He visits his trusted friend and partner in crime, Syphon, with a hope to find good fighters. Syphon has been building many alien robots, so that they can be used at any time of crisis. He locked up the robots, one in each cell, in his laboratory. As he had already tasted defeat on earth by Max years ago, this time Vilgax wanted to be careful with the crew he would hire. He wanted to know how many of the robots were correctly programmed and are capable of think cleverly and fight for themselves. When told, Syphon come up with an idea to find the intelligent robots among them.

In Syphon's laboratory, 100 robots are locked in 100 separate cells, numbered 1 to 100. He gives one coin to each robot. He then goes to each cell, in ascending order, with a bowl. Each robot can decide whether to put a single coin in the bowl, depending on its current content. After collecting from the 100th room, if the bowl is empty, the game ends, and nothing happens (Vilgax wouldn't take any of the robots as his army). Otherwise, Syphon shows his bowl to the robots who haven't contributed any coins. These robots can guess from which cells those coins are, with the following consequences:

1. A wrong guess leads to immediate disqualification of the robot, Syphon will dismantle them.
2. A correct guess leads to immediate qualification of the robot, Vilgax recruits it in his army.
3. If the robots refuse to guess, nothing happens.
4. If there's no correct guess, all coin-givers will be taken into the army.

All coins are identical. And Syphon has programmed all the robots in such a way that they prefer life to freedom, and freedom to imprisonment, and will choose to guess if and only if they're 100% sure (they don't take chances with their own lives, but would love to see others dismantled). Each robot knows its own cell number and is locked in its cell throughout the game. There's no information exchange of any kind between the robots. All the above are common knowledge.



How many robots will be freed and hired by Vilgax?

- A. Between 95-100(including both)
- B. Between 90-94(including both)
- C. Between 85 - 89 (including both)
- D. Less than and equal to 84

17) 'The Plumbers' get an intelligence report that Vilgax is recruiting an army of fierce alliance from eight different planets to reclaim the Omnitrix and destroy Earth. So, max asks some (eight) of his old team members to try to gather more information about Vilgax's evil alien army. The eight of them target one planet each and start working on their highly efficient supercomputers. After an hour of (searching), they all manage to retrieve some confidential/secret info about their targeted planet. But before sending the collected information to Max and other plumbers, they decide to share the info about all the planets among themselves in order to cross-check. They want to share the info through a series of bilateral conversations(e.g., via telephone). Devise an efficient(in terms of the total number of conversations) algorithm for this task. Assume that in every conversation, both parties exchange all the info they know at the time.

In how many minimum no. of conversations can these agents exchange all the information among themselves?

INTEGER QUESTION

SEC - 7:

Fighting Vilgax

Marking Scheme - Shield me

You get 5 marks for answering correctly and 4 marks are deducted for answering incorrectly, but if you have solved the previous question correctly, then there is no negative marking for the current question.

For example:

If Your sequence is RWRW, then your score is $5+0+5+0 = 10$

But if your sequence is RWWR, then your score would be $5+0-4+ 5 = 6$

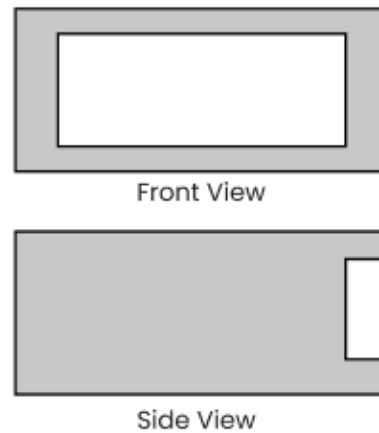
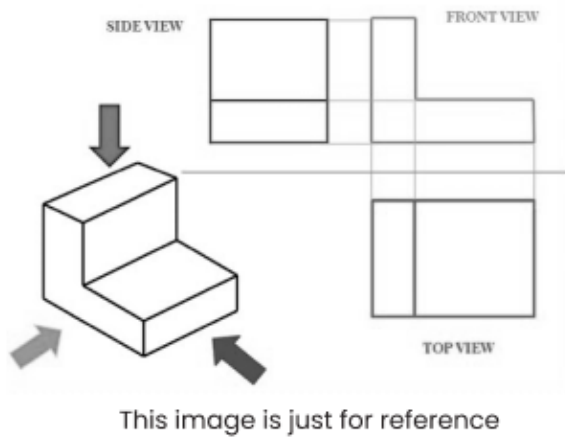
And if your sequence is RWUW, then your score is $5+0+0-4 = 1$

Where (R -> right answer, W-> wrong answer, U-> Unattempted)

Maximum marks : 15

Minimum marks : -12

18) Vilgax sends his men to destroy the secret base of the plumbers. Max notices a strange and heavily armed vehicle coming towards the base and warns the agents. But they couldn't figure out what the vehicle was, so one of the agents got a few images of the vehicle from a nearby security camera and showed it to the max. The top figure shows the front view, and the below figure shows the side view of the vehicle.



Max suggests that if they could fix a bomb on every edge of the vehicle, then it would definitely be destroyed before it could reach them. The challenge they had now was to figure out the shape of the vehicle and find the number of edges it has so as to fix a bomb on every edge.

Edge, here, means - In the case of straight planes as in a cube, the meaning would be the same, i.e., it would have 12 edges. But for curved surfaces, like a simple cone, the no. of edges is 1, and for a simple cylinder, it is 2.

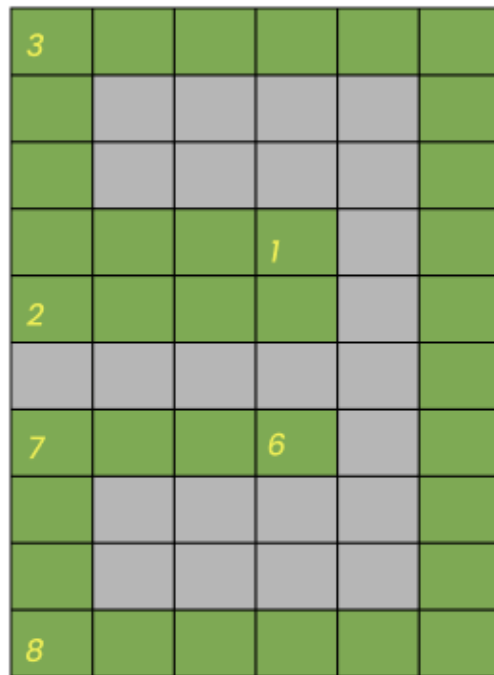
Gwen comes to their rescue and figures out the shape of the vehicle. Ben has to go to the vehicle and fix a bomb on every edge. Ben pulls his hand up, rotates the dial of the Omnitrix and turns into XLR8 and rushes to the vehicle. Assuming that the vehicle has the simplest possible three-dimensional shape, find the minimum number of bombs ben would need to take with him.

INTEGER QUESTION



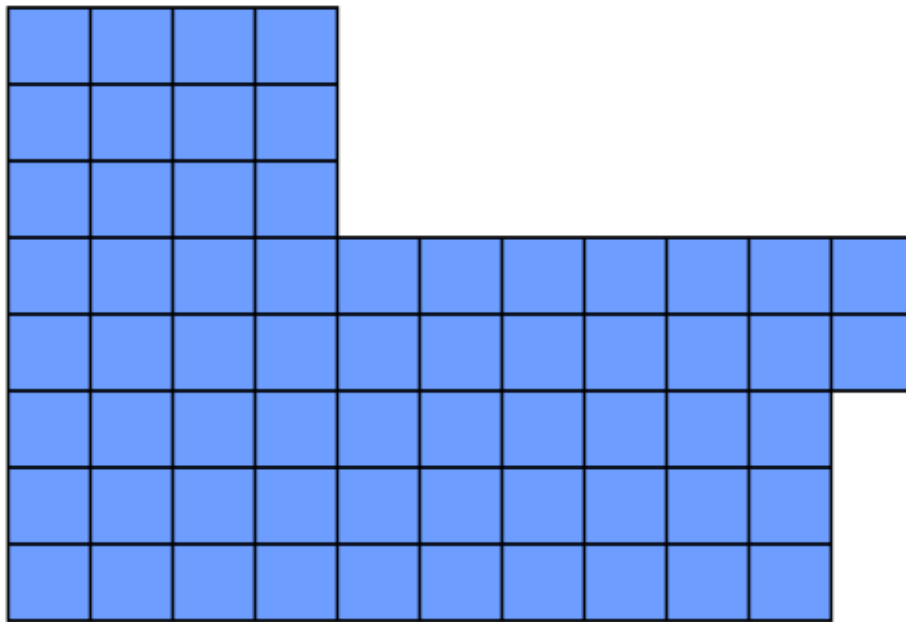
19) The plumbers' satellites spotted that Vilgax's spaceship has entered the realm of Earth. Ben, Gwen and Max reach the site and sneak into Vilgax's spacecraft. But some of Vilgax's crew spots them and start attacking them. Ben transforms into Four-Arms and starts smacking the enemy while Max and Gwen use the weaponry brought from the Plumbers' base to fight back, but their efforts were going in vain as Vilgax had a very big alien army. To put an end to this, Ben decides to face Vilgax himself. He goes into the heart of the spaceship to duel with him. When Ben comes, Vilgax laughs at him that he isn't a worthy opponent to dual with him, both physically and intellectually. But seeing FourArms starts shaking with rage, Vilgax challenges him to first come to him through his chamber and then dual with him.

Vilgax's chamber has a specially designed floor, in the form of a 10x6 grid as shown in the image below. Vilgax tells Ben he can choose to come from any corner of the chamber. If he chooses corner 1, he can start from any square labelled from 1 and 3 (both inclusive); and if he chooses to start from corner 2, he can start from any square labelled from 4 to 6 (both inclusive). The one at corner 1 gets to move first. Vilgax and Ben have to move towards each other with every step, taking turns. Both of them can move either by one square or two, but not more than two, and they shouldn't jump over each other at any point. They can move only in **green** boxes and not in **red** squares. In order to block the moves of Vilgax and win over him, which corner should Ben begin to move and in what minimum moves can Ben corner Vilgax?



- A. Player 1
- B. Player 2
- C. Draws at both
- D. More information required

20) After the game, Vilgax and Fourarms come out of the chamber and start dualling furiously. While fighting they both fall on the floor with a thud and the nearby cabin breaks open and many metal pieces fall down. Max finds a familiar, thick plate with small squares on it. The figure is given below.



He remembers that when this plate is cut into two pieces and fixed correctly, it forms a rectangle which becomes 'The Key', that he can use to let the ship Self-destruct. Max asks Ben and Gwen to construct the 'Key' while he's engaged in fighting with Vilgax. He instructs them that the cutting should be done only along the sides of the small squares but not through them. Gwen figures out and marks a line which needs to be cut. Ben turns into Diamondhead and cuts the plate along the marked line with his crystals.

He fixes the so obtained pieces and gets a rectangle of $m \times n$ dimensions. Gwen distracts Vilgax, while Ben passes 'The Key' to Max, who immediately activates the ship's self-destruct and sets the timer to 30 seconds. Diamondhead holds Gwen and Max and jumps out of the ship, leaving Vilgax in it as it explodes. Find the square of the length of the diagonal in exact dimensions if the length of the side of smaller squares is 1 unit.

INTEGER QUESTION

Section 8

21. Diamondhead holds Gwen and Max and jumps out of the ship, leaving Vilgax in it as it explodes.

Thus, Ben, along with Max and Gwen, saves the Earth from Vilgax and his evil alien army.

But his vacation isn't over yet, and he's very much looking forward for more such adventures. Max shows them the map and asks them to choose a place for their following camping site. Ben and Gwen excitedly spin the pointer on the map, and the pointer stops on East India...on Guwahati!

The three Tennysons start their journey to the campus of IIT Guwahati as many more adventures await for them during the MAINS Event of Technothon'22.

On a scale of 1 to 5, how was your experience of technothon?

INTEGER QUESTION